Pruning Attention Heads the Way the Brain Prunes Vessels

Anonymous submission

Abstract

Transformers keep more attention heads than their tasks need, and pruning methods decide how many to keep by hand. The brain sizes its blood supply without a target: vessels that carry little flow regress, and Murray's law sets vessel size by trading the work of pumping blood against its upkeep. We propose *hemodynamic attenuation*, a pruning rule built on this principle. Each attention head pays a fixed upkeep, is valued after the other heads compensate for it, and closes gradually when it is worth less than its upkeep. Because the rule minimizes a single cost, loss plus upkeep, what it keeps can be predicted before training, and one setting, an upkeep of 1% of the loss the heads remove, worked on both the benchmark and GPT-2 without a target size. On a new task whose minimal number of heads is known, it kept exactly that number in 29 of 30 runs while the task stayed solved, which none of the published methods we tested achieved. Trained from scratch to copy repeated text, it kept the heads of the known induction circuit in 10 of 10 runs. On GPT-2 small, it recovered the core of the hand-mapped indirect-object circuit, stable across prompt samples: with 20–21 heads instead of 26, it matched the published circuit's accuracy and favored the correct name more strongly.

Introduction

A trained transformer usually has more attention heads than its task needs (Michel, Levy, and Neubig 2019; Voita et al. 2019), and trained models keep backup heads that take over when others are removed (Wang et al. 2023; McGrath et al. 2023). Existing pruning methods fix the final size by hand, as a target sparsity (Zhu and Gupta 2017), a penalty strength (Voita et al. 2019; Nam et al. 2025) or a threshold (Conmy et al. 2023), and most prune after training. None of them predicts the size it will reach or how much redundancy it will keep (Table 1). And benchmarks with known circuits build the circuit in by construction (Lindner et al. 2023; Gupta et al. 2024), so they cannot test whether a method finds the minimal circuit among the redundant heads that training produces.

The cerebral circulation solves a similar allocation problem without a target size. The neurovascular unit senses local demand and adjusts vessel diameter (Iadecola 2017). Vessels that carry little flow regress gradually rather than being cut (Korn and Augustin 2015), and tissue survives the loss of a vessel when collaterals supply its territory. Murray (1926) showed that vessel size follows from minimizing the work of moving blood plus the metabolic cost of keeping it. By the same logic, a vessel is worth keeping only while it earns its upkeep.

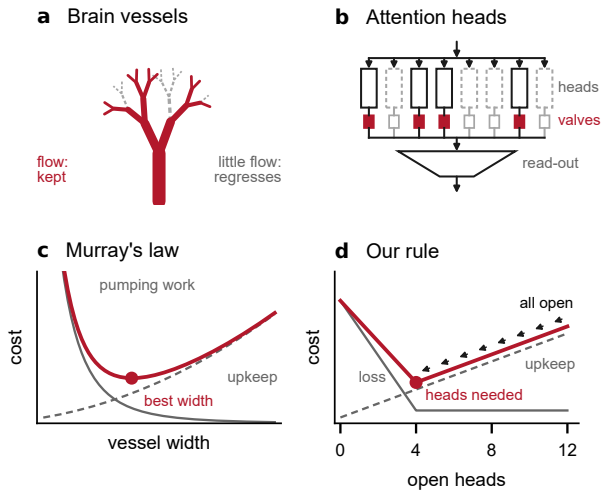


Figure 1: From vessels to heads (illustration). **(a)** Brain vessels that carry little flow regress (dashed), and the rest are kept. **(b)** Each attention head passes through a valve. Heads worth less than their upkeep are closed (gray). **(c)** Murray's law: a vessel's width minimizes the work of pumping blood plus the upkeep of the blood it holds. **(d)** Our rule: loss falls until the needed heads are open and upkeep grows with each one, so closing heads lowers the sum until only the needed heads remain. A design principle, not a model of blood flow.

We carry this principle over to attention heads (Figure 1). Our contributions are:

- *Hemodynamic attenuation*, a rule that charges each head a fixed upkeep, values it after the other heads compensate, and closes heads gradually during training. It minimizes loss plus upkeep (Proposition 1), so it makes predictions

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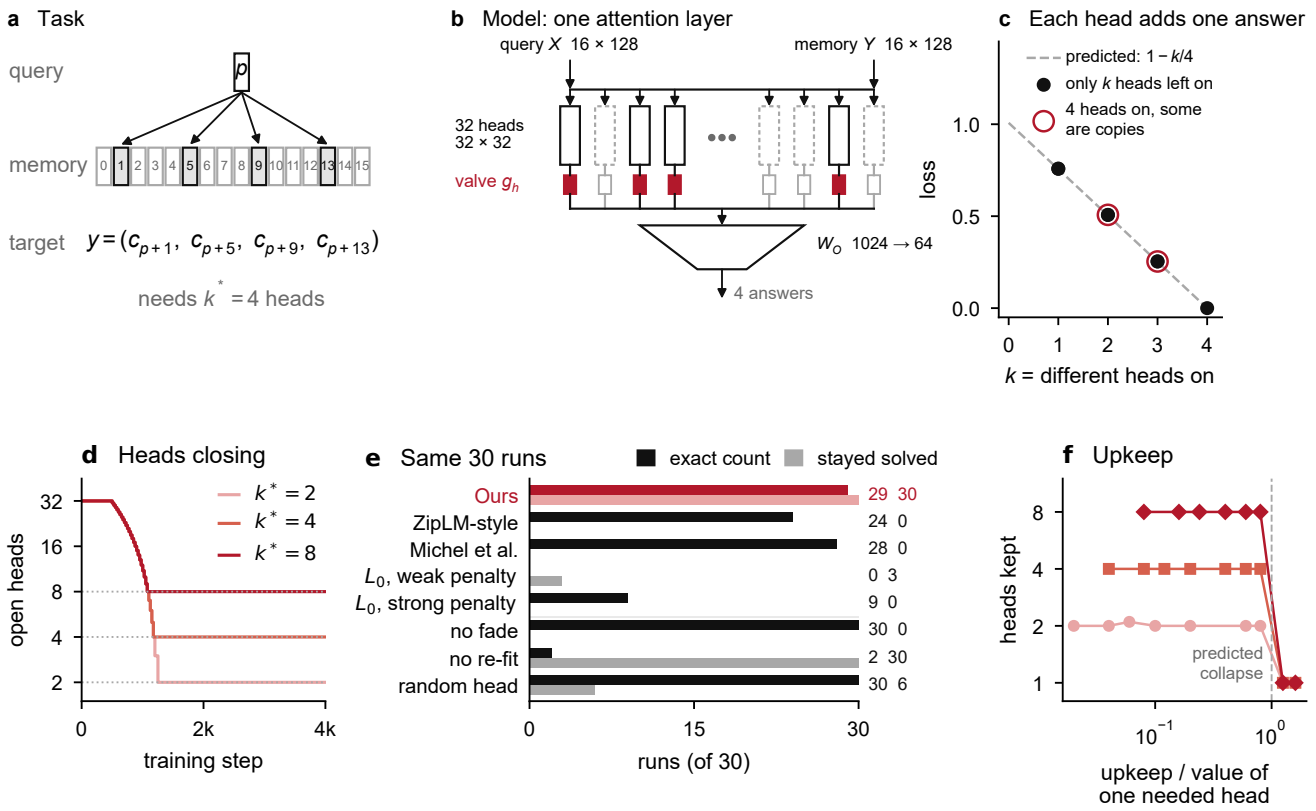


Figure 2: A task with a known minimal circuit, and the results on it. **(a)** A query names a position p, and the target is the items at R fixed offsets. Here $R = 4$, so 4 heads are needed. **(b)** The model, with sizes. Each head has a valve g_h (red when open) and pays an upkeep while open. **(c)** Loss of a trained model with only k heads on (black) follows $1 - k/R$ (dashed). With copies among the heads (red), only *different* heads count: a copy is worth nothing. Units of L_{triv}. **(d)** Open heads during training, one run per task size, with $k^\star$ dotted. **(e)** Every method on the same 30 runs ($k^\star = 2, 4, 8$, 10 seeds each), all run in our code (Kurtic, Frantar, and Alistarh 2023; Michel, Levy, and Neubig 2019; Voita et al. 2019). ZipLM uses our score once, then fine-tunes. ZipLM and Michel et al. were given the true count. L_0 gates are shown at the weakest and strongest of three penalties ($\lambda = 0.05$ and 1). The lower three rows each remove one part of our rule. **(f)** Heads kept against the upkeep, in units of the value of one needed head, L_{triv}/R. Theory predicts $k^\star$ heads left of 1 and one head right of it (3 seeds per point).

before training.

- A shared benchmark for AI and neuroscience: its minimal circuit is known in size and roles, while the redundancy comes from training.
- Three predictions of the objective confirmed: the head count, the upkeep at which the rule collapses, and the redundancy kept under failure.
- Known circuits recovered: induction, and a stable core of the indirect-object circuit in GPT-2 small that matches the published circuit with fewer heads. Heads hidden by their backups are found by re-valuing and missed by ranking once.

Hemodynamic Attenuation

Valves. Each head's output passes through a valve $g_h \in [0, 1]$ before the heads are summed, $\hat{T} = \sum_h g_h O_h W_O^h$. A head with $g_h = 0$ adds nothing and receives no gradient, so a model with k open valves is exactly a k-head model.

Collateral value. For the set S of open heads, let $L^*(S) = \min_{W_O} L$ be the loss after the output weights of the heads in S are re-fitted. The value of an open head is $v_h = L^*(S \setminus h) - L^*(S)$ and the gain of a closed head is $L^*(S) - L^*(S \cup h)$, both in closed form from the heads' output covariance. This is the Optimal Brain Surgeon score (Hassibi and Stork 1993) for whole heads, as in ZipLM (Kurtic, Frantar, and Alistarh 2023). If two heads output the same signal, the survivor takes over its twin's weight and the loss does not change, so a copy is worth nothing.

Schedule. All valves start open. After a warm-up of 10% of training, every 25 steps the rule closes the lowest-valued open head if $v_h < \pi$, and otherwise reopens the highest-gain closed head if its gain exceeds 2π. A closing valve falls by $1/100$ per step, so a head regresses while the others adapt.

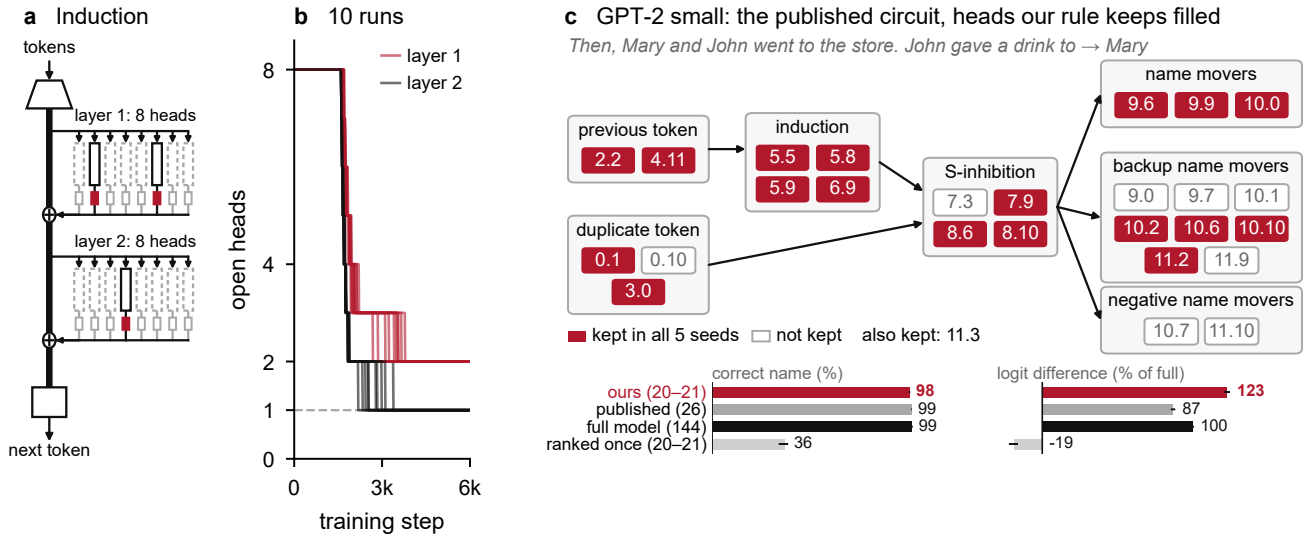


Figure 3: Known circuits. **(a)** The induction model: two attention-only layers of 8 heads on a residual stream, each head with a valve. **(b)** Open heads per layer during training, 10 runs. Layer 2 closes to the one induction head. **(c)** GPT-2 small on indirect object identification, frozen weights, prompts of Wang et al. (2023). The 26 heads of the published circuit are grouped by the roles they assigned, from early to late layers, and a head (layer.head) is filled if our rule keeps it at the lowest upkeep in all 5 seeds. The rule assigns no roles. Bottom: held-out accuracy and logit difference (as % of the full model), ranges over seeds. *Ranked once*: the same number of heads, chosen by their single removal from the full model.

Method	Unit	During training	Re-fit score	Re-values each step	No target size	Predicts size
Ours	head	✓	✓	✓	✓	✓
CHG	head	–	–	–	✓	–
ZipLM	head	–	✓	✓	–	–
Michel	head	–	–	✓	–	–
Voita	head	✓	–	–	✓	–
ACDC	edge	–	–	✓	✓	–
RigL	weight	✓	–	✓	–	–
OBS	weight	–	✓	✓	–	–

Table 1: Related methods: causal head gating (CHG) (Nam et al. 2025), ZipLM (Kurtic, Frantar, and Alistarh 2023), Michel, Levy, and Neubig (2019), Voita et al. (2019), ACDC (Conmy et al. 2023), RigL (Evci et al. 2020) and Optimal Brain Surgeon (OBS) (Hassibi and Stork 1993). *Re-fit*: a unit is valued after the others compensate. *No target size*: no sparsity or count is given (a penalty or threshold still sets it). *Predicts size*: the size reached can be computed before running, where the value of a needed head is known (here, the benchmark).

These settings were not tuned. Varying the fade (25–400 steps), warm-up (5–25%) or check interval (10–50 steps) kept the exact count in 18 of 18 runs ($k^\star = 4$) and the task solved in 17. The upkeep is $\pi = 0.03\, L_{\text{triv}}$, where L_{triv} is the loss of predicting the mean.

The rule optimizes an objective. Let $J(S) = L^*(S) + \pi|S|$, the loss plus the upkeep of the open heads, the form of Murray's cost (Figure 1d).

Proposition 1. *Closing h lowers J exactly when $v_h < \pi$, and reopening a head whose gain exceeds 2π lowers J by more than π. At a fixed point, no single removal lowers J and no single addition lowers it by more than π. If, in every set of open heads, each needed head is worth more than π and each redundant head less, then from all heads open the rule stops at exactly the needed heads, for any such π.*

The first two claims follow from $J(S \setminus h) - J(S) = v_h - \pi$ and $J(S \cup h) - J(S) = \pi - \text{gain}_h$. Since every step lowers J, the rule stops. It is forward–backward greedy selection (Pudil, Novovičová, and Kittler 1994; Zhang 2011) over groups of weights, run while the heads are still learning.

A Benchmark with a Known Minimal Circuit

The task adapts sparse retrieval (Sanford, Hsu, and Telgarsky 2023) so that the number of heads, not the width, is the constraint. A memory of $N = 16$ slots holds random items c_j, redrawn for every example. A query names a position p, and the target is the R items at fixed offsets, $(c_{p+\delta_1}, \ldots, c_{p+\delta_R})$ (Figure 2a). A head returns one weighted average of the items, one linear combination of the R unknowns, so R items need R heads with different attention patterns. With $k \le R$ such heads the lowest achievable loss is

$$L^*(k) = (1 - k/R)\, L_{\text{triv}}, \qquad (1)$$

confirmed on trained models to within $0.003\, L_{\text{triv}}$ (Figure 2c). Each needed head is worth L_{triv}/R and each copy 0, so Proposition 1 predicts exactly $k^\star = R$ heads for any $\pi < L_{\text{triv}}/R$, and above it a collapse to the single head the

rule never closes. The model is one cross-attention layer with $H = 32$ heads, width 128 and no MLP, trained with AdamW (learning rate 2×10^{-3}, batch 128, 4000 steps). The task is solved when the loss is below $0.02\, L_{\mathrm{triv}}$. *Exact count* means the open heads over the last 40% of training equal $k^\star$. *Stayed solved* means that once solved, the loss never rises above the bar again.

Results

Exact count while the task stays solved. The rule kept exactly $k^\star$ heads in 29 of 30 runs, and the loss never left the solved range (Figure 2d,e). No published method did both. Michel et al. and ZipLM, given the true count, lost the solution during pruning in every run: the loss rose to a median of 0.52 and $0.38\, L_{\mathrm{triv}}$ before further training recovered it, and all runs ended solved. L_0 gates trade one failure for the other: a weak penalty kept the loss near the bar but never reached the count, and a strong one reached it in 9 runs but left 26 of 30 unsolved at the end. Without the fade the solution was lost, as with a vessel cut rather than regressed, and without the re-fit the rule valued copies and kept too many heads.

The upkeep does not set the size, until theory says it should. Every upkeep from 0.01 to $0.1\, L_{\mathrm{triv}}$ gave the exact count. Above $\pi = L_{\mathrm{triv}}/R$ every needed head costs more than it saves, so the rule should close down to one head. All 18 runs below this point kept exactly $k^\star$ heads and all 18 above it kept one (Figure 2f), with the final loss Equation (1) gives for one head.

Changing demand. When the heads needed went from 2 to 4 and back to 2 during training, the rule closed back to 2 in 10 of 10 runs but reached 4 in only 2, because a closed head that never learned the new items has no value to earn reopening. Adding *sprouting*, which opens the most valuable closed head while the loss stays unsolved, as oxygen-starved tissue grows new vessels, tracked 2, 4 and 2 heads in 10 of 10 runs.

A third prediction: redundancy under failure. The objective also says how much redundancy to keep. If each head fails independently with probability p at every training step, the value of the B-th head is $v(B) = \frac{1-p}{R}\Pr[\mathrm{Bin}(B-1, 1-p) < R]\, L_{\mathrm{triv}}$: it pays only when it works and fewer than R of the others do, a k-out-of-n calculation from reliability theory (Barlow and Proschan 1965), and the rule should keep the largest B with $v(B) \geq \pi$. There is no free parameter. We committed predictions for 100 new runs before starting them, over $k^\star = 2, 4, 8$, p from 0.05 to 0.3 and two upkeeps. 79 matched exactly and all 100 were within one head. Keeping $k^\star$ heads, or enough that $k^\star$ work on average, would match 4 and about 33.

Induction. A two-layer attention-only model with $8 + 8$ heads was trained to continue a repeated sequence. Its known circuit is a previous-token head in layer 1 and an induction head in layer 2 (Olsson et al. 2022). Cross-entropy has no closed-form re-fit, so here the lowest-valued head is closed only if the loss without it stays below $0.05\, L_{\mathrm{triv}}$, and the last closed head reopens if the loss exceeds twice that. The rule kept $2 + 1$ heads with the expected roles in 10 of 10 runs, and the task stayed solved (Figure 3a,b). The second layer-1 head acts as a backup, cutting errors on the repeat from 1 in 8 to 1 in 200.

GPT-2 small. We applied the rule to the 144 heads of GPT-2 small on indirect object identification (Figure 3c), whose 26-head circuit was found by hand (Wang et al. 2023). The weights were frozen, so this tests the valuing and re-valuing alone, without fade or reopening. Each of 5 seeds draws 300 prompts to choose heads and 1500 held-out prompts from the generator of Wang et al. (2023). A closed head is replaced by its mean output on prompts with three different names, and the upkeep is a fraction (0.05, 0.02, 0.01) of the loss gap between all heads closed and all open. At 1% of the gap, the same scale as $0.01\, L_{\mathrm{triv}}$ on the benchmark, where it kept the exact count in 9 of 9 runs, the rule kept 20–21 heads and picked the correct name in 98–99% of held-out prompts, matching the published 26 heads (99.0–99.4%) and the full model (99.1–99.8%) with 5–6 fewer heads. They are the right heads: 19 of them are in the published circuit, 19 were kept in all 5 seeds, the name movers in all 15 runs, and the negative name movers, which push against the correct name, in none. Higher upkeeps gave nested subsets of 10–11 and 17–18 heads in every seed, so the upkeep trims the circuit rather than changing it. Re-valuing matters. The same number of heads ranked once by their removal from the full model picked the correct name in only 33–38% of prompts, below chance. Ranking once misses early heads whose value is hidden while their backups are present. On the logit difference, the measure of Wang et al. (2023), the 20–21 heads recovered 121–124% of the full model's, against 85–88% for the published circuit, consistent with closing the negative name movers. The kept heads are a circuit for one task, not a smaller general model: alone they fail on greater-than and repeated tokens. Closing them cut repeated tokens from 83% to 11–12% (random sets of the same size: 76–78%), because the induction heads kept here are the ones that task relies on.

Discussion

Hemodynamic attenuation turns a principle of the cerebral circulation into a pruning rule with a stated cost. Like the vessels it is modeled on, it adapts the model's size to demand during training. Because the cost is explicit, what it does can be worked out in advance. The number of heads, the upkeep at which it collapses and the redundancy it keeps under failure each follow from the objective, and each was confirmed. In GPT-2 small, the same rule found a smaller, stable core of a circuit mapped by hand. The models trained from scratch are small so that their circuits are known, GPT-2 small was frozen, and the proposition covers fixed heads. Next, we will apply the rule while training a language model.

What it predicts for the brain. The correspondence in Figure 1 runs both ways. The same redundancy calculation would give the number of collateral vessels worth keeping from how often a vessel is occluded and what it costs to maintain: more where occlusion is likelier or tissue costlier to lose, none where upkeep exceeds the expected loss, a prediction that can be tested.

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